Module 3: Operators in C

In C programming, **operators** are like tools. They help you **do things with numbers, letters, or values**.

Think of it like math in school, but on a computer.

3.1. Arithmetic Operators (Math Operators)

These are used to **do calculations**.

Operator	What it does	Example
`+`	Add two numbers	`3 + 2 = 5`
`-`	Subtract	`5 - 2 = 3`
`*`	Multiply	`3 * 2 = 6`
`/`	Divide	`6 / 2 = 3`
`%`	Remainder (mod)	`5 % 2 = 1`
`++`	Add 1 (increment)	`a++` adds 1 to a
`--`	Subtract 1 (decrement)	`a--` subtracts 1

Example in C:

```
#include <stdio.h>
int main() {
    int a = 5, b = 2;
    printf("a + b = %d\n", a + b); // 7
    printf("a - b = %d\n", a - b); // 3
    printf("a * b = %d\n", a * b); // 10
    printf("a / b = %d\n", a / b); // 2
    printf("a %% b = %d\n", a % b); // 1
    a++;
    printf("a after increment = %d\n", a); // 6
}
```

Mini Exercise:

1. Calculate 10 + 15 - 5 × 2 and print it in C.
2. Increase a number by 1 using `++`.

3.2. Relational Operators (Comparison Operators)

These are used to **compare numbers**. They tell the computer if something is true or false.

Operator	Meaning	Example
`==`	Equal to	`a == b`
`!=`	Not equal	`a != b`
`>`	Greater than	`a > b`
`<`	Less than	`a < b`
`>=`	Greater or equal	`a >= b`
`<=`	Less or equal	`a <= b`

Example:

```
#include <stdio.h>
int main() {
    int a = 5, b = 8;
    printf("a == b? %d\n", a == b); // 0 (false)
    printf("a != b? %d\n", a != b); // 1 (true)
    printf("a < b? %d\n", a < b);   // 1 (true)
}
```

Mini Exercise:

1. Check if 10 is greater than 5.
2. Check if two numbers are equal.

3.3. Logical Operators (Combine Conditions)

Think of it like asking: "Is this true **and** that true?" or "Is this true **or** that true?"

Operator	Meaning	Example
`&&`	AND	Both must be true
`		`
`!`	NOT	Reverse the truth

Example:

```
#include <stdio.h>
int main() {
    int a = 5, b = 10;
    if (a > 0 && b > 0)
        printf("Both numbers are positive!\n");
    if (a > 0 || b < 0)
        printf("At least one number is positive!\n");
    if (!(a > b))
        printf("a is not bigger than b!\n");
}
```

Mini Exercise:

1. Check if a number is **positive AND even**.
2. Check if a number is **negative OR zero**.

3.4. Assignment Operators (Give a Value to a Variable)

These are like **shortcuts to update numbers**.

Operator	Meaning	Example
`=`	Assign value	`a = 5`
`+=`	Add and assign	`a += 3` → a = a + 3
`-=`	Subtract and assign	`a -= 2`
`*=`	Multiply and assign	`a *= 2`
`/=`	Divide and assign	`a /= 2`
`%=`	Modulus and assign	`a %= 3`

Example:

```
#include <stdio.h>
int main() {
    int a = 5;
    a += 3; // a = 8
    a *= 2; // a = 16
    printf("a = %d\n", a);
}
```

Mini Exercise:

1. Start with a number 10, add 5, multiply by 2, subtract 4, and print the result.

3.5. Bitwise Operators (Operate on Bits)

Every number in a computer is stored in **binary** (0s and 1s). Bitwise operators work **on these 0s and 1s**.

Operator	Meaning	Example
`&`	AND	`0101 & 0011 = 0001`
`	`	OR
`^`	XOR	`0101 ^ 0011 = 0110`
`~`	NOT	`~0101 = 1010`
`<<`	Left shift	`0101 << 1 = 1010`
`>>`	Right shift	`0101 >> 1 = 0010`

Example:

```
#include <stdio.h>
int main() {
    int a = 5, b = 3;
    printf("a & b = %d\n", a & b); // 1
    printf("a | b = %d\n", a | b); // 7
}
```

Mini Exercise:

1. Check if a number is even using &.
2. Multiply a number by 2 using <<.

3.6. Ternary Operator (Shortcut for if-else)

Instead of writing if-else, you can do it in **one line**:

```
condition ? value_if_true : value_if_false;
```

Example:

```
#include <stdio.h>
int main() {
    int a = 10, b = 20;
    int max = (a > b) ? a : b;
    printf("Maximum = %d\n", max);
}
```

Mini Exercise:

1. Find the **smaller number** between two numbers using ternary operator.

3.7. Increment / Decrement Operators

- **Prefix:** `++a` → Add 1 **before** using the number
- **Postfix:** `a++` → Add 1 **after** using the number

Example:

```
#include <stdio.h>
int main() {
    int a = 5, b;
    b = ++a; // a=6, b=6
    b = a++; // a=7, b=6
}
```

Mini Exercise:

1. Predict the output:

```
int x = 5;
printf("%d %d", x++, ++x);
```

Tip for Remembering Operators:

- **Arithmetic:** + - * / % ++ -- → Math stuff

- **Relational:** == != > < >= <= → Comparing
- **Logical:** && || ! → True/False logic
- **Assignment:** = += -= *= /= %= → Give and update values
- **Bitwise:** & | ^ ~ << >> → Work on binary
- **Ternary:** ? : → Quick if-else
